Thrangborwell Sohlang

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Summary

AI/ML Engineer and Technical Solutions Engineer with a foundation in Computational Physics (M.Sc., IIT Hyderabad), experienced in production-grade document AI, Generative AI, client-facing technical delivery, and computer-vision systems. Skilled in Python, SQL, FastAPI, RAG, LangChain, OpenAI, Claude, and production workflow automation, with experience shipping client-used features, reporting dashboards, TMS integrations, and document-processing business rules.

Skills

- **Programming:** Python, SQL
- **ML/AI:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, YOLO (v3–v8), DeepSORT, VGG
- **LLMs and Agents:** OpenAI API, Claude, Claude Code, RAG, LangChain, LangGraph, CrewAI, AutoGen, prompt engineering
- **APIs and Tooling:** FastAPI, Pydantic, OpenAPI/Swagger, Postman
- **Data and Infra:** pandas, BigQuery, Scrapy, Selenium, Git, Docker, Linux, REST APIs
- **Document AI and Integrations:** PaperEntry, CargoWise, Descartes, NocoDB, SFTP, XML, JSON, document AI validation, business rules
- **Vector Databases:** Pinecone, ChromaDB, FAISS, Qdrant
- **Cloud:** AWS (Bedrock, S3)
- **Deployment and UI:** FastAPI, React, Streamlit, VLLM, Ollama
- **Optimisation:** ONNX, TensorRT, OpenVINO
- **Testing:** PyTest, unit testing

Experience

Technical Solutions Engineer - Deep Cognition AI — Remote

Nov 2025 – Present

- Shipped production features for PaperEntry used by logistics and customs clients, including reporting dashboards, investigation tools, export workflows, shipment-processing utilities, and TMS integrations with CargoWise and Descartes.
- Built visual reporting dashboards with filters, sorting, investigation views, and exports to PDF, HTML, and XLS to help clients monitor shipment and document-processing performance.
- Worked directly with clients through meetings, live issue reviews, and requirement discussions, resolving production issues and converting feedback into product improvements.
- Investigated document-processing issues and refined business rules for extraction correction, value propagation, decimal handling, product-code mapping, invoice fallback logic, and supplier-specific behaviour.
- **Tech:** Python, pandas, SQL, regex, Linux, Docker, SFTP, XML, JSON, NocoDB, CargoWise, Descartes, PaperEntry, OpenAI, Claude, Claude Code, document AI validation, business rules

AI Engineer - Sustainability Economics — Bangalore

Mar 2025 – May 2025

- Appraised 3 document-AI platforms (Google Document AI, AWS Data Automation, Azure Document Intelligence) for structured extraction from PDFs and spreadsheets.
- Established an AI-driven ETL prototype on AWS with CloudWatch monitoring and lineage logging, covering ingest, extract and load stages end-to-end.
- Realised 92% field-level extraction accuracy on a labelled pilot of 20 documents; compiled an error taxonomy and remediation playbooks.
- Co-ordinated multi-agent workflows using 3 frameworks (LangGraph, CrewAI, AutoGen) to automate multi-step data operations.
- **Tech:** AWS (S3, Bedrock), Google Document AI, Azure Document Intelligence, LangGraph, CrewAI, AutoGen, Python, Docker

AI/ML Developer - Sarjen Systems (Client of DataLabs) — Ahmedabad

Aug 2024 – Feb 2025

- Upgraded 1 enterprise RAG system for document search across PDFs and flowcharts: delivered 500 ms median query latency and improved search accuracy by 60%.
- Indexed 1,000 documents with custom parsers to normalise graph and flow artefacts for structured retrieval and ranking.
- Engineered 1 real-time golf-ball tracking proof-of-concept (YOLO + DeepSORT) with 60 s end-to-end latency on 30 s, 120 FPS footage.
- **Tech:** LangChain, custom parsers, YOLO, DeepSORT, Python

Data Scientist Associate - DataLabs AI Hyderabad
Jul 2024 – Feb 2025

- Constructed 1 Scrapy + Selenium pipeline that processed 20 GB of marketing data with resilient crawl and retry logic.
- Cut manual cleaning time by 45% through normalisation and outlier filtering across the processed dataset.
- Introduced 1 prototype GPT assistant for structured sales-data querying to accelerate ad-hoc analysis.
- **Tech:** Scrapy, Selenium, Python, SQL

Data Science Intern - Response Informatics Ltd. Hyderabad
Oct 2023 – Jun 2024

- Co-developed 1 NLP dashboard that generated SQL from natural language and rendered live visualisations in Streamlit.
- **Tech:** Streamlit, SQL, Python, NLP

Projects (Top 3)

HTS Agent (Harmonised Tariff Schedule) - Hybrid Search & Duty Calculator
AI Agent 2025

- Engineered hybrid retrieval (BM25 + embeddings) over HS codes, chapter and section notes, and rulings with transparent citations; achieved 20 ms average time-to-answer.
- Devised a rule-aware duty calculator applying chapter and section notes plus tariff formulae; exposed FastAPI endpoints with unit tests and guardrails.
- Reduced policy look-up errors by 80% and consolidated multi-step look-ups into 1 query-and-compute flow.
- **Tech:** LangChain, ChromaDB, FastAPI, Python, Docker, embeddings, BM25, OpenAPI/Swagger

GPT-Powered Data Visualisation Tool 2024
Internal Tool

- Enabled natural-language SQL and chart generation using GPT-4 and LangChain across a schema with over 20 tables for business stakeholders.
- Enhanced quality controls with evaluator checks, retry logic, and error handling, shortening the query-to-report cycle by over 60%.
- **Tech:** OpenAI API, LangChain, SQL, Streamlit

Computational Physics & Research Experience

Gamma-Ray Burst Afterglow Model Comparison — IIT Hyderabad

- Compared simple and broken power-law models for GRB050525a afterglow using frequentist model selection.
- Applied chi-square, AIC, and BIC to evaluate goodness of fit; broken power law achieved approximately 40% lower chi-square, capturing the observed jet break.
- Computed likelihoods using custom Python implementation with NumPy, SciPy, and Matplotlib.
- Demonstrated end-to-end statistical modelling, data fitting, and model comparison.

Education

M.Sc., Physics - Indian Institute of Technology Hyderabad (IIT Hyderabad)
Aug 2020 – Aug 2022

- Compared 2 afterglow models using AIC/BIC; broken power-law outperformed single power-law on the evaluated dataset.
- Coursework included 2 advanced modules: Computational Physics; Data Science for Astronomy.

Awards and Languages

- **Scholarship:** INSPIRE Scholarship for Higher Education (2017–2022).
- **Languages:** Khasi (native), English (fluent).